Sequences, Series & Complex Numbers

ClipSAT Printable Practice Suite (clipsat.org) • Topic 3.01

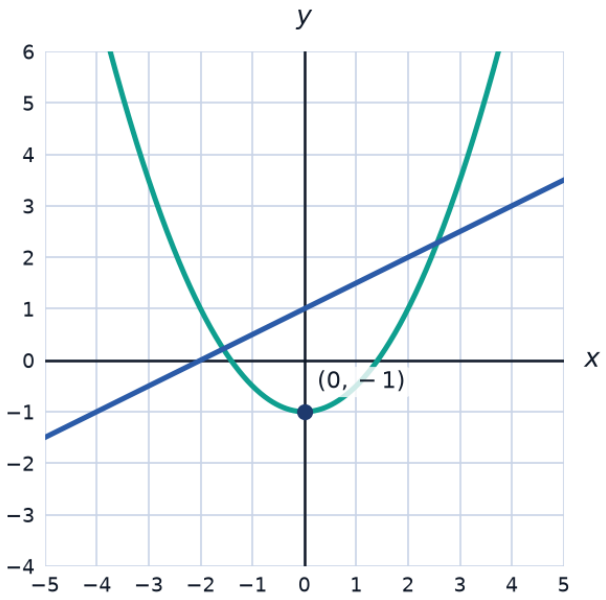
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Arithmetic sequences

Find the 12th term of the arithmetic sequence $5, 9, 13, \ldots$

2 [Foundational] • Target: Arithmetic sequences

The 4th term of an arithmetic sequence is 18, and the 9th term is 38. Find the first term and the common difference.



3 [Foundational] • Target: Arithmetic sequences

Find the sum of the first 20 terms of the arithmetic sequence $7, 11, 15, \ldots$

4 [Foundational] • Target: Arithmetic sequences

Find the number of terms in the arithmetic sequence $6, 10, 14, \ldots, 150$.

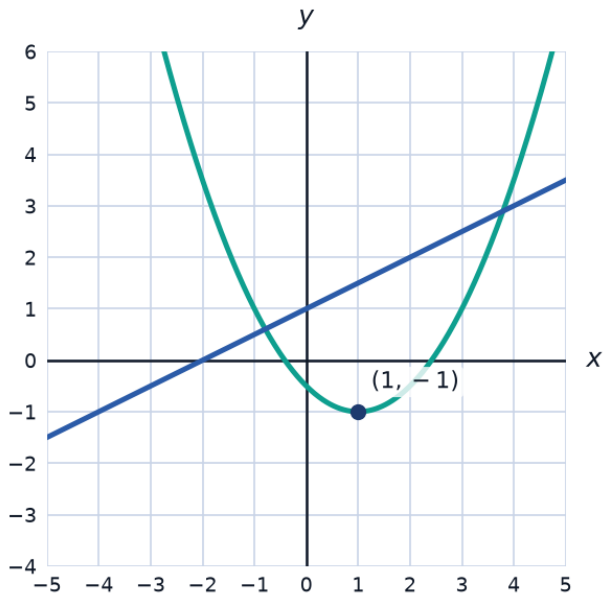
5 [Foundational] • Target: Geometric sequences

Find the 7th term of the geometric sequence $3, 6, 12, \ldots$

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6 [Foundational] • Target: Geometric sequences

In a geometric sequence with all positive terms, the 3rd term is 20 and the 5th term is 80. Find r and a_1.



7 [Foundational] • Target: Geometric sequences

Given that $x - 1$, $x + 1$, and $3x - 1$ are consecutive terms of a geometric sequence, find the possible value(s) of x.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Geometric sequences

Find the sum of the first 6 terms of the geometric sequence $5, 15, 45, \ldots$

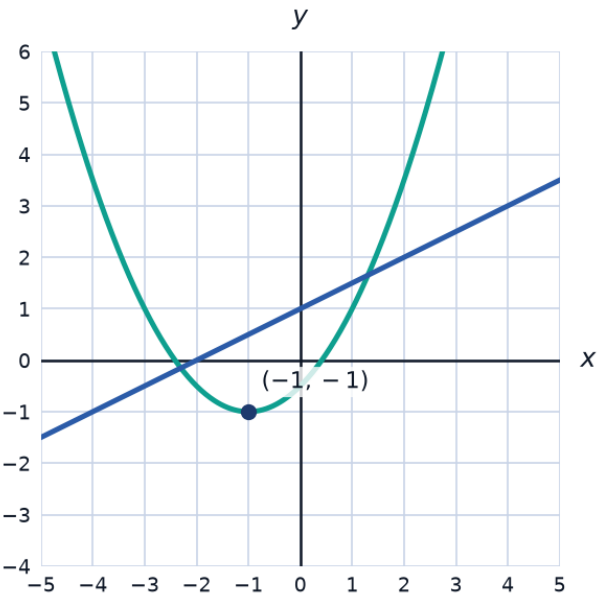
9 [Intermediate] • Target: Series and sigma notation

Evaluate $\sum_{k=1}^{5} (3k - 2)$.

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10 [Intermediate] • Target: Series and sigma notation

Write the series $3 + 7 + 11 + \cdots + 39$ in sigma notation, then evaluate it.



11 [Intermediate] • Target: Series and sigma notation

Evaluate $\sum_{n=1}^{6} 2^n$.

12 [Intermediate] • Target: Geometric series

Find the sum of the first 8 terms of the geometric sequence $2, 6, 18, \ldots$

13 [Intermediate] • Target: Geometric series

Find the sum to infinity of $16 - 8 + 4 - 2 + \cdots$

14 [Intermediate] • Target: Geometric series

Express the repeating decimal $0.4\overline{5}$ (that is, $0.45555\ldots$) as a geometric series and find its exact fractional value.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

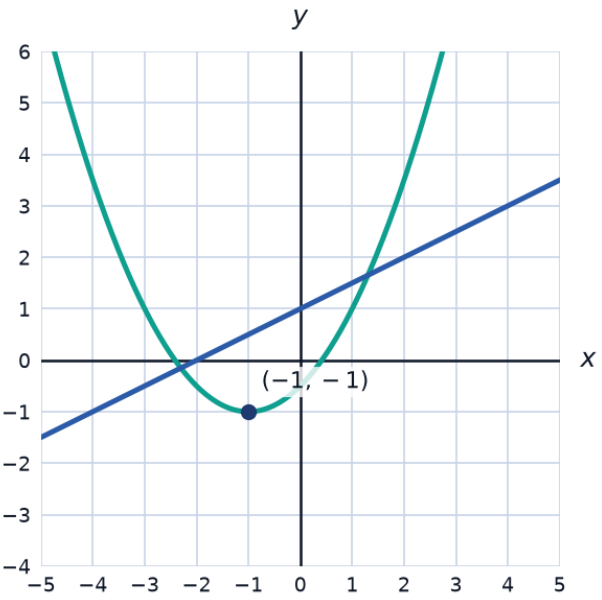
15 [Challenging] • Target: Complex number arithmetic • Cross-Lesson: 4.01 Lines, Triangles & Polygons

Given $z_1 = 5 - 2i$ and $z_2 = -3 + 4i$, find $z_1 + z_2$ and $z_1 - z_2$.

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16 [Challenging] • Target: Complex number arithmetic • Cross-Lesson: 5.01 Trigonometry

Multiply $(3 + 2i)(4 - 5i)$.



17 [Challenging] • Target: Complex number arithmetic • Cross-Lesson: 6.01 Circles & Coordinate Geometry

Simplify i^{23}, using $i^2 = -1$.

18 [Challenging] • Target: Complex number arithmetic • Cross-Lesson: 7.01 Solid Geometry & Similarity

Simplify $(2 - 3i) - (5 + i) + (1 - 4i)$.

19 [Challenging] • Target: Complex conjugates and division • Cross-Lesson: 8.01 Statistics & Probability

Find the conjugate of $z = 6 + 7i$, and compute $z\bar{z}$.

20 [Challenging] • Target: Complex conjugates and division • Cross-Lesson: 1.01 Advanced Algebra

Simplify $\frac{4 + 3i}{1 - 2i}$ by multiplying by the conjugate of the denominator.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Arithmetic sequences

Find the 10th term of the arithmetic sequence $10, 16, 22, \ldots$

a. 66

b. 70

c. 64

d. 58

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22 [Foundational] • Target: Geometric sequences

Find the 14th term of the arithmetic sequence $3, 5, 7, \ldots$

a. 29

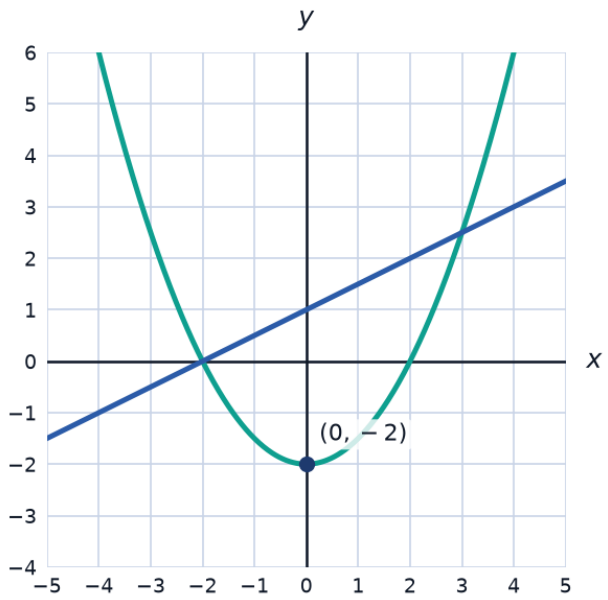
b. 31

c. 27

d. 32

23 [Foundational] • Target: Series and sigma notation

Find the 11th term of the arithmetic sequence $-3, -6, -9, \ldots$



a. -36

b. -30

c. -31

d. -33

24 [Foundational] • Target: Geometric series

Find the 13th term of the arithmetic sequence $-13, -7, -1, \ldots$

a. 53

b. 65

c. 61

d. 59

25 [Foundational] • Target: Complex number arithmetic

Find the 15th term of the arithmetic sequence $12, 8, 4, \ldots$

a. -40

b. -48

c. -44

d. -42

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26 [Foundational] • Target: Complex conjugates and division

Find the 10th term of the arithmetic sequence $11, 17, 23, \ldots$

a. 59

b. 67

c. 65

d. 71

27 [Foundational] • Target: Modulus of complex numbers

Find the 10th term of the arithmetic sequence $11, 14, 17, \ldots$

a. 35

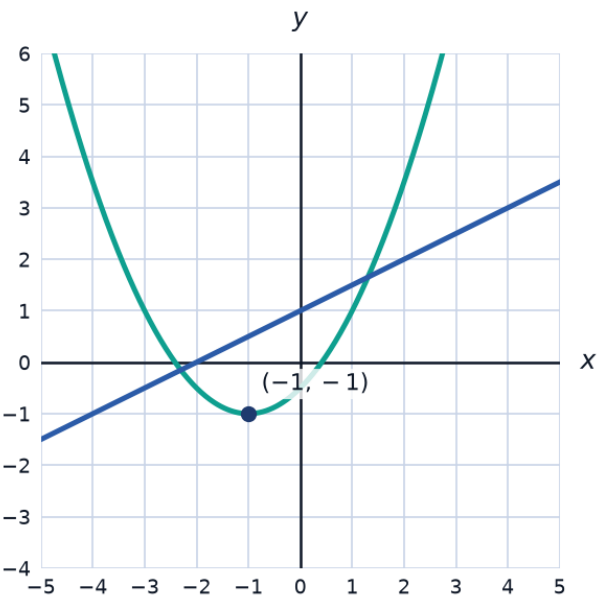
b. 38

c. 40

d. 41

28 [Foundational] • Target: Quadratic equations with complex roots

Find the 4th term of the geometric sequence $-1, 2, -4, \ldots$



a. 6

b. 8

c. 10

d. -8

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: A well-designed word problem: a bouncing ball

Find the 6th term of the geometric sequence $-6,\ -12,\ -24, \ldots$

a. -192

b. -190

c. 192

d. -194

30 [Intermediate] • Target: Arithmetic & Geometric Progressions

Find the 7th term of the geometric sequence $-8,\ -24,\ -72, \ldots$

a. -5830

b. 5832

c. -5834

d. -5832

31 [Intermediate] • Target: Binomial Expansions & General Terms

Find the 3rd term of the geometric sequence $7,\ 21,\ 63, \ldots$

a. 63

b. 61

c. -63

d. 65

32 [Intermediate] • Target: Operations in the Complex Plane

Find the 6th term of the geometric sequence $8,\ -24,\ 72, \ldots$

a. -1946

b. -1944

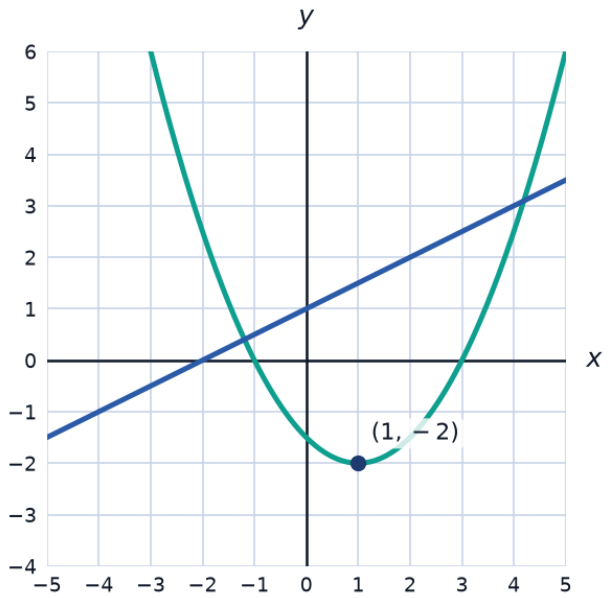
c. -1942

d. 1944

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33 **[Intermediate]** • **Target: De Moivre's Formula & Roots**

Find the 3rd term of the geometric sequence $-6,\ 12,\ -24, \ldots$



a. 24

b. -26

c. -24

d. -22

34 **[Intermediate]** • **Target: Arithmetic sequences**

Find the 4th term of the geometric sequence $-1,\ -2,\ -4, \ldots$

a. -10

b. 8

c. -8

d. -6

35 **[Intermediate]** • **Target: Geometric sequences**

Find the sum of the first 4 terms of the geometric sequence $-9,\ -18,\ -36, \ldots$

a. -135

b. -133

c. -144

d. -137

36 **[Intermediate]** • **Target: Series and sigma notation**

Find the sum of the first 7 terms of the geometric sequence $-4,\ 12,\ -36, \ldots$

a. -2188

b. -2192

c. -2190

d. -2186

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37 [Intermediate] • Target: Geometric series

Multiply $(2 - 7i)(5 + 3i)$.

a. $31 - 27i$

b. $31 - 29i$

c. $31 + 41i$

d. $-11 - 29i$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Complex number arithmetic • Cross-Lesson: 7.01 Solid Geometry & Similarity

Multiply $(5 + 5i)(-6 - 4i)$.

a. $-50 - 50i$

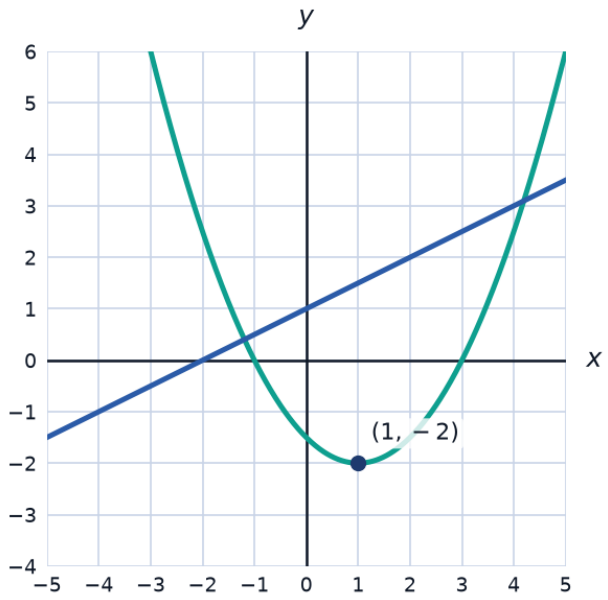
b. $-10 - 48i$

c. $-10 - 50i$

d. $-10 + 10i$

39 [Challenging] • Target: Complex conjugates and division • Cross-Lesson: 8.01 Statistics & Probability

Find $|z|$ for $z = 6 - 8i$.



a. 8

b. 10

c. 14

d. 12

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40 [Challenging] • Target: Modulus of complex numbers • Cross-Lesson: 1.01 Advanced Algebra

Find $|z|$ for $z = -11 + 60i$.

a. 59

b. 61

c. 71

d. 63

41 [Challenging] • Target: Arithmetic & Geometric Progressions • Cross-Lesson: 2.01 Functions: Composite, Inverse & Logarithmic

In an arithmetic progression, the sum of the first n terms is given by $S_n = 3n^2 + 2n$. What is the 10th term a_{10}?

a. 59

b. 62

c. 65

d. 320

42 [Challenging] • Target: Arithmetic & Geometric Progressions • Cross-Lesson: 4.01 Lines, Triangles & Polygons

Evaluate the sum of the convergent geometric series: $\sum_{n=0}^{\infty} 5(-\frac{1}{3})^n$.

a. $\frac{15}{4}$

b. $\frac{15}{2}$

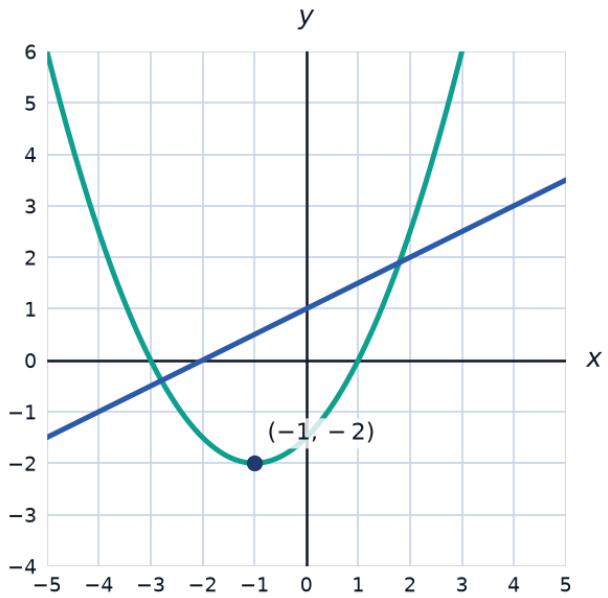
c. $\frac{5}{4}$

d. 5

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43 [Challenging] • Target: Binomial Expansions & General Terms • Cross-Lesson: 5.01 Trigonometry

What is the sum of the coefficients in the expansion of $(5x - 4)^7$?



a. 1

b. 128

c. 2187

d. 0

44 [Challenging] • Target: Operations in the Complex Plane • Cross-Lesson: 6.01 Circles & Coordinate Geometry

What is the modulus of the complex number $z = (3 + 4i)(5 - 12i)$?

a. 65

b. 17

c. 13

d. 25

45 [Challenging] • Target: De Moivre's Formula & Roots • Cross-Lesson: 7.01 Solid Geometry & Similarity

Evaluate $(\cos 30^\circ + i\sin 30^\circ)^6$.

a. -1

b. 1

c. i

d. $-i$

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